PORTFOLIO / 2026 / VERSION 1.0 AQUA / LAP

Waley El-Nagdi

Software Developer

Backend Development & Systems Integration

Python · REST APIs · PostgreSQL · Business Applications

A self-taught developer applying for a company placement through the AQUA pathway toward the Lehrabschlussprüfung (LAP) in Applikationsentwicklung - Coding.

PERSONAL DETAILS

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FEATURED PROJECT

NexusOS

MULTI-TENANT PLATFORM

React SPA on an async FastAPI service layer with a PostgreSQL tenant model: authentication, RBAC, rate limiting and company-scoped data isolation.

DOMAIN nexusos.com
GITHUB github.com/waley7337
HOSTING Frontend on Vercel / Backend on Railway

PYTHON · FASTAPI · POSTGRESQL · REDIS · REACT · TYPESCRIPT · DOCKER

5	7	~110	31
ARCHITECTURE LAYERS	SECURITY LAYERS	SECURITY TESTS	UI COMPONENTS

PROJECT SUMMARY, LINKS & DEPLOYMENT: see following pages

REDACTED COPY - date of birth and age removed; links to the private NexusOS repository replaced with the public GitHub profile. The original document is unchanged and not published.

01 / PROJECT SUMMARY

NexusOS

A multi-tenant business platform where authentication, authorization and tenant isolation were built as infrastructure — before any product module was allowed to depend on them.

OVERVIEW

NexusOS is a web platform designed to host multiple business capabilities — CRM, projects, finance, HR, inventory — behind one application shell, one identity model and one tenant boundary. Rather than shipping a module first and retrofitting multi-tenancy later, the identity, permission and isolation layers were solved first so that everything built on top inherits them.

The platform layer is complete and tested end to end. The product modules above it are, with the dashboard interface as the exception, not yet built.

MY ROLE

Sole developer
System architecture, backend, frontend, database design, security implementation, design system, test suite and CI pipeline.

LINKS & DEPLOYMENT

DOMAIN
nexusos.com

GITHUB
github.com/waley7337

FRONTEND DEPLOYMENT
Vercel

BACKEND DEPLOYMENT
Railway

CLOUDFLARE

External security perimeter · Primary integration
This layer sits in front of the application and protects it at the network and HTTP edge, before requests reach the backend.

INTENDED SECURITY FUNCTIONS

WAF and DDoS protection
Filtering malicious traffic and mitigating denial-of-service attacks.

Bot protection and traffic challenges
Detecting or challenging suspicious automated requests.

TLS and trusted-proxy handling
Protecting traffic in transit and ensuring the backend trusts client IP forwarding only from explicitly trusted infrastructure.

KEY DEVELOPMENT HIGHLIGHTS

Tenant scope is derived, never accepted.
Authenticated requests carry no tenant header, path segment or body field. Company identity is read from the caller's stored record on every request, so a valid token for one tenant cannot address another.

Authorization is re-derived from the database, not read from token claims.
Role and permissions are loaded fresh on every protected request and fail closed on suspended users or deactivated tenants — so revocation takes effect on the next request, not at token expiry.

Refresh-token reuse is treated as compromise.
Refresh tokens rotate against server-side session records; replaying a spent token revokes every session for that user and writes an audit event.

Isolation is enforced twice.
Application-level scoping plus composite foreign keys — (id, company_id) — that make a cross-tenant relationship unrepresentable in the database, independent of application correctness.

Layered request security.
CORS allowlisting with production startup validation, security headers, Redis-backed rate limiting on credential endpoints, and a trusted-proxy boundary so IP-based controls cannot be spoofed by a forwarded header.

A design system built as infrastructure.
157 design tokens and 31 components with a living style-guide route, plus a single-flight token refresh in the API client so concurrent 401s don't invalidate each other through rotation.

TECHNOLOGY

AREA	TECHNOLOGIES
Frontend	React 19, TypeScript, Vite, React Router, TanStack Query, Zustand, React Hook Form, Zod, Tailwind CSS, Radix UI, Recharts
Backend	Python 3.12, FastAPI, Pydantic v2, SQLAlchemy 2.0 async, Alembic, Argon2, PyJWT, structlog
Data	PostgreSQL 16 via asyncpg, Redis 7 for rate-limit counters
Infrastructure & testing	Docker Compose, GitHub Actions CI, pytest against live PostgreSQL and Redis, Vitest with Testing Library, ruff, mypy

STATUS

IMPLEMENTED

Multi-tenant identity model, full authentication lifecycle, RBAC with database re-validation, the security middleware layer, and the frontend platform including the design system. 167 backend and 89 frontend test functions, with CI running lint, type checks, migrations and both suites.

PREPARED

38 domain model classes migrated for CRM, projects, finance, HR and inventory with no endpoints yet; a tenant-scoped repository implemented and tested but not yet on live routes; the dashboard interface complete against fixture data.

NOT IMPLEMENTED

The product modules behind the sidebar, AI features (tables modelled, no inference code), and PostgreSQL row-level security. Tenant isolation is currently application-level and relational.